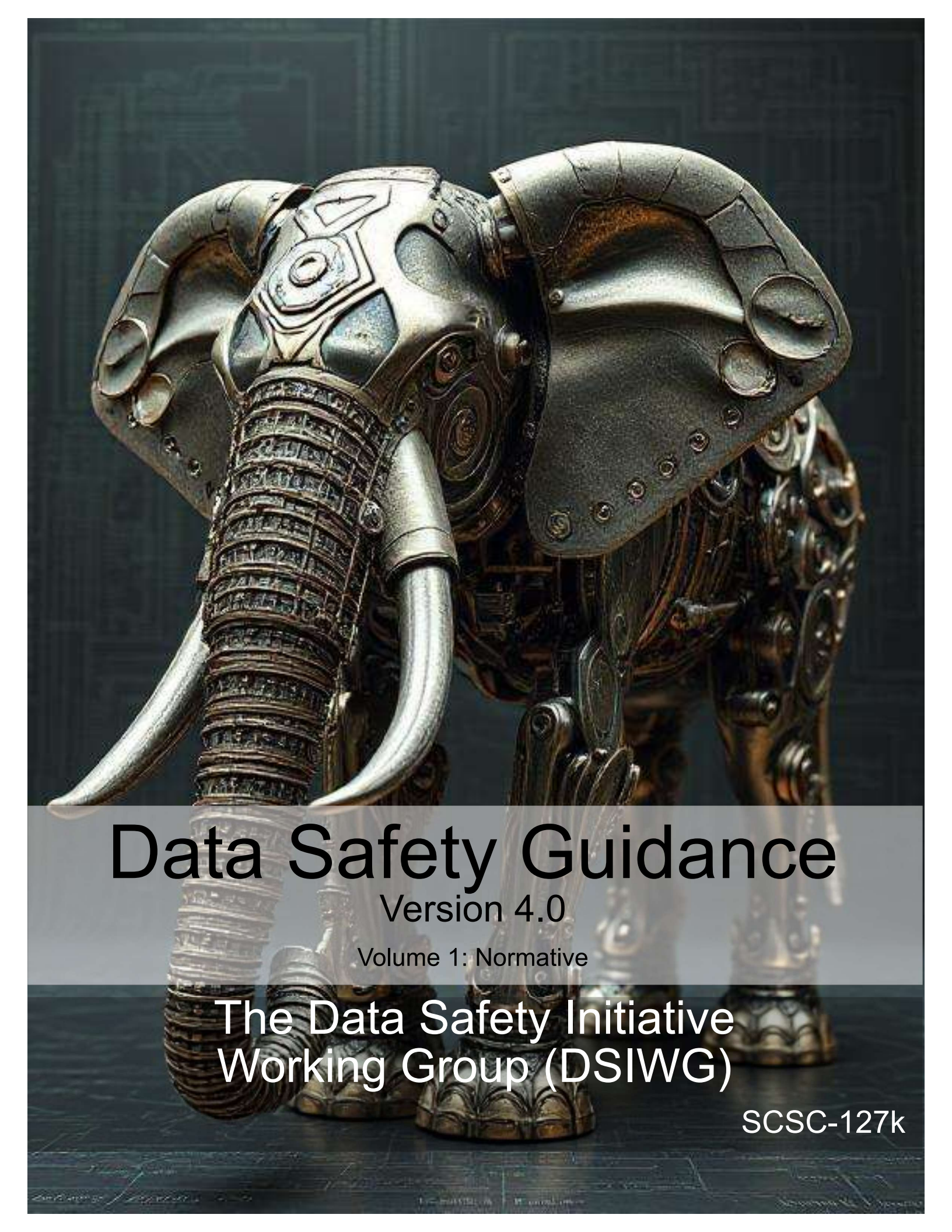




# Data Safety Guidance

Version 4.0

Volume 1: Normative

The Data Safety Initiative  
Working Group (DSIWG)

SCSC-127k



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The [Safety-Critical Systems Club \(SCSC\)](#) is the professional network for sharing knowledge regarding safety-critical systems. It brings together:

- engineers and specialists from a range of disciplines working on safety-critical systems in a wide variety of industries;
- academics researching the arena of safety-critical systems;
- providers of the tools and services that are needed to develop the systems; and
- the regulators who oversee safety.

Through publications, seminars, workshops, tutorials, a web site and, most importantly, at the annual [Safety-critical Systems Symposium \(SSS\)](#), it provides opportunities for these people to network and benefit from each other's experience in working hard at the accidents that don't happen. It focuses on current and emerging practices in safety engineering, software engineering, and product and process safety standards.

This document was written by the [Data Safety Initiative Working Group \(DSIWG\)](#), which is convened under the auspices of the [SCSC](#). The document supports the [DSIWG's](#) vision, which is to have clear guidance that reflects emerging best practice on how data (as distinct from software and hardware) should be managed in a safety-related context. This update takes account of the consensus that a process-based guidance document will complement existing safety management processes, making it more usable. It was formally released at [SSS'26](#), 10–12 February 2026, details of which may be found at <https://scsc.uk/events-diary/safety-critical-systems-club-symposia/safety-critical-systems-symposium-2026>

Comments on this document are actively encouraged. These can be emailed to:

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# Data Safety Guidance

The Data Safety Initiative Working Group [DSIWG]

February 2026

## Change History

| Version | By                             | Status                                                | Date        |
|---------|--------------------------------|-------------------------------------------------------|-------------|
| 1.0     | The <a href="#">DSIWG</a> Team | First draft for external review                       | 31-JAN-2014 |
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| 1.3     | The <a href="#">DSIWG</a> Team | For publication at <a href="#">SSS'16</a>             | 29-JAN-2016 |
| 2.0     | The <a href="#">DSIWG</a> Team | For publication at <a href="#">SSS'17</a>             | 30-JAN-2017 |
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| 3.3     | The <a href="#">DSIWG</a> Team | For publication at <a href="#">SSS'21</a>             | 09-FEB-2021 |
| 3.4     | The <a href="#">DSIWG</a> Team | For publication at <a href="#">SSS'22</a>             | 08-FEB-2022 |
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| 3.7     | The <a href="#">DSIWG</a> Team | For publication at <a href="#">SSS'25</a>             | 04-FEB-2025 |
| 4.0     | The <a href="#">DSIWG</a> Team | For publication at <a href="#">SSS'26</a>             | 10-FEB-2026 |

### Changes Since the Last Edition

This is a major rewrite of the guidance. The document has been split into three volumes , each giving progressively more detail.

The process remains substantially the same. The intention is that compliance with this version does imply compliance with earlier versions [???Need to check this is still true](#), but some material that was informative in earlier versions has been made normative, so compliance with earlier versions of the guidance does not guarantee compliance with this version.

### Future work

[???Update future work](#) [???Consider whether future work needs to be included in all parts](#) MCA Ltd has continued to work with the [DSIWG](#) to develop a prototype software tool to assist in the automation of the processes described in this guidance document. A working version of the tool has been developed and organizations that could benefit from the use and further development of the tool are urged to contact MCA at [Mission Critical Applications Limited \(radish@mca-ltd.com\)](mailto:radish@mca-ltd.com).

A number of improvements to the guidance are currently planned. These improvements are intended to clarify the application of the data safety process [???Review future work](#) and include:

- further detail on the assurance of communications and data flows,
- data safety considerations associated with distributed [datasets](#) and Blockchain,
- addition of new [treatments](#) to the tables in [??](#),
- review of the tables of [treatments](#), with the aim of making them easier to use,

- further explanation of some [treatments](#), where their use or benefit is not immediately apparent,
- reordering of parts of the document to improve readability, especially as regards likelihood,
- further detail on tool assurance,
- harmonisation of language and guidance on how organizations may expand the tables to incorporate their own internal processes.
- guidance on the application of the data safety culture questionnaire,

Several of these changes are likely to cause parts of the document to be re-ordered – they have therefore been deferred to the next major update, in version 4.0 of the guidance.

If you or your organization are interested in learning more about the work of the [DSIWG](#) or joining any of the sub-groups, please visit the [SCSC](#) website, where more information including contact details may be found on the “Working groups” section of the site.

## Related working groups

The [SCSC](#) sponsors initiatives to develop methods and techniques through a number of working groups. These groups each address safety aspects peculiar to their domain, including data aspects when appropriate. The current list of working groups includes:

- Assurance Cases,
- Security Informed Safety,
- Safe AI Working Group,
- Safer Complex Systems.



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## Foreword

*Data is here. Data is growing. Data is causing harm.*

**Data is here:** Data is becoming ever more important in our lives: influencing, managing and even controlling many critical aspects. The use of [artificial intelligence \(AI\)](#) systems is a new, exciting, but potentially hazardous use of data. [Large language model \(LLM\)](#) based systems are trained on vast amounts of data, and it is this data which enables them to be useful.

Also true  
of ML.  
Mention  
that?

Some of this data is related to our personal safety and well-being. Consider, for example, the importance of data defining the layout of railway signals, data that indicates the position of underwater obstructions in nautical channels, or data that is used to train a vision recognition system to detect tumours in medical images. Organizations now make significant decisions (including safety-related decisions) based solely on data held in systems. Hence, organizations need to safely manage, control and process their data. In particular, they must actively manage key [data properties](#) that preserve safety.

**Data is growing:** There are at least two reasons why the use of data has grown and, equally important, why it is expected to continue to grow. The first relates to the rapid expansion of the area loosely termed “Big Data”, including the use of large [datasets](#) to support machine learning and [AI](#) applications. The second is the growing use of systems of systems, where data is the lifeblood that connects together disparate elements and allows a cohesive capability to be built. Put simply, the need to address data-related issues is a pressing problem and will continue to be.

**Data is causing harm:** Strictly speaking, data can neither cause nor prevent harm. However, mistakes in data or inappropriate uses of data within safety-related systems have been factors in a number of documented accidents and incidents. Examples include aircraft attempting to take off from the wrong runway (and consequently crashing), ships running aground, and patients being exposed to higher than planned doses of radiation.

Against this background, the [DSIWG](#) was established under the auspices of the [SCSC](#). The [DSIWG](#)’s aim is to develop clear, cross-sector guidance that reflects emerging best practice on how data (as opposed to software or hardware) should be managed in a safety-related context. For the most part, this guidance is based on well-established techniques, and it has been designed to be compatible with current safety standards and to integrate with existing safety management systems. What is new, however, is the explicit and relentless focus on data, making it a “first-class citizen” within system safety analyses. Because of this focus, this guidance should help organizations identify, analyse, evaluate and treat data-related risks, thus reducing the likelihood of data-related issues causing harm in the future.

## Quick Start Guide

*Data really powers everything that we do.*  
**Jeff Weiner**

This section provides a single-page introduction to [data safety guidance \(DSG\)](#). For first-time readers this should help place individual sections within an appropriate context. It should also help returning readers quickly navigate the document's contents.

When reference is made to the content of other volumes, this is indicated by the volume number shown in bold text. So “Chapter **3:5**” refers to chapter 5 in volume 3.

Systems are changing. The role of data is becoming more prominent. Hence, data needs to be considered as a “first-class citizen” in system safety analyses. This will help mitigate organizational and system-level risks associated with the use of data.

The guidance is in three volumes, published as separate documents.

**Volume 1** (this volume) contains the normative material. It describes the “what” of the guidance and states what is needed to ensure conformance with this guidance, if required. It should be of use to those establishing processes and procedures for data safety assurance and for those doing the actual assurance;

**Volume 2** contains the informative material. It describes the “how” of the guidance and should be of particular value to data safety practitioners;

**Volume 3** contains the discursive material. It describes the “why” of the guidance and gives background and supporting material intended to improve practitioner's understanding of the data safety management processes and of issues relating to data safety management.

A data safety management process has been developed. This is based on four phases:

- establish context;
- identify risks;
- analyse risks; and
- evaluate and [treat](#) risks.

???This part needs rewriting when structure is stable

- The underlying principles of the guidance are described in [Chapter 3](#).
- The objectives and outputs to satisfy the principles are described in Chapter ??.
- Definitions and abbreviations are described in [Chapter 4](#).
- The activities of each phase (and associated [tailoring information](#)) are described in Chapter **2:??**.
- A worked example is provided in Chapter 7.
- Guidance for the assignment of [data safety assurance levels \(DSALs\)](#) is given in Chapter **2:??**.

- Examples of ways that DSALs may be customised, with particular focus on likelihood are given in Chapter 2:??.
- Issues relating to AI and machine learning (ML) are considered in Chapter 2:?? and Chapter 2:??.
- Additional information on data categories is given in Chapter 2:??.
- Issues that may arise when migrating, porting, importing or exporting data are discussed in Chapter 2:??).
- Lifecycle considerations are addressed in Chapter 3:??.
- The data safety tool RADISH is described in Chapter 3:??.
- Some of the data issues that made management of the Covid-19 virus difficult are discussed in Chapter 3:??.
- A summary of accidents and incidents in which data was potentially a causal factor is given in Chapter 3:??;

We never seem to explain what we mean by data categories. Consider doing so.

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# 1 Introduction

???consider whether the introduction needs to be included in all parts (normative, informative and discursive, perhaps and shall, should, may, but they should be in definitions anyway)

*We're entering a new world in which data may be more important than software.*

**Tim O'Reilly**

## 1.1 Aim and Scope

This guidance document aims to:

- describe the data safety problem;
- provide methods for identifying and analysing levels of risk; and
- recommend methods and approaches for evaluating and treating those risks.

It has been written for a wide readership. Its target audience is all those who have an interest in or a responsibility for safety-related data within systems, including managers, developers, safety engineers, assurers (including independent safety auditors), regulators, and operators.

The document is also intended to cover a number of different sectors. It identifies a wide spectrum of safety-related data that exists in many forms within systems, from specification and requirements data to maintenance and disposal data, and everything in between. In particular, this document is not just concerned with numerical or well-structured data used during system operation.

The contents of the document are considered mature enough to be useful, but represent current thoughts on what is a complex and evolving area. Furthermore, to allow it to be produced within a reasonable timescale, this edition focuses on key items. It is not intended to be exhaustive. For example, this guidance document does not consider issues relating to staff competence or organizational structure.

There are various understandings of the word “[safety](#)”, from those limited to physical harm to the person: “The state of being protected from or guarded against hurt or injury; freedom from danger” ([Oxford English Dictionary \(OED\)](#)) to those that consider more general harm: “Freedom from conditions that can cause death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.” ([1]). This guidance concentrates on the former but is applicable more widely, and from time to time other types of harm will be considered, particularly but not exclusively in the examples of incidents and accidents (Chapter **3:5**).

## 1.2 Structure of the Guidance Documents

The guidance is in three parts, published as separate documents.

**Part 1** contains the normative material. It describes the “what” of the guidance and states what is needed to ensure conformance with this guidance, if required. It should be of use to those establishing processes and procedures for data safety assurance and for those doing the actual assurance work;

**Part 2** contains the informative material. It describes the “how” of the guidance and should be of particular value to data safety practitioners;

**Part 3** contains the discursive material. It describes the “why” of the guidance and gives background and supporting material intended to improve practitioner’s understanding of the data safety management processes and of issues relating to data safety management.

### 1.3 Intended Relationship to Other Documents

This document is intended to be used as a supplement to existing standards and norms that are relevant to the scope of the work being undertaken. It may be used to provide a deeper insight into the risks that data poses to the project team’s outputs, allowing them to produce credible improvements to the safety argument. Where a standard or norm sets out specific data-related objectives then, unless agreed otherwise with the regulator or safety duty holder, they shall take precedence over the guidance provided herein.

In the longer term, the hope is that future standards and norms will take up relevant concepts, approaches and methods from those in this document. The [DSIWG](#) also hopes that organizations will include the concepts, approaches and methods in their own safety management processes.

## 1.4 Normative, Informative and Discursive Text

Three types of text are used within this guidance document:

**Normative** text, which is prescriptive. Typically, this text is restricted to describing objectives and outputs.

**Informative** text, which is descriptive text that is closely linked to the normative text. Typically, this text provides a suggested way by which conformance with the normative text may be achieved, but alternative means of conformance are possible.

**Discursive** text, which contains discussions that are relevant to the general topic of data safety, but which are not closely linked to the normative text. A discussion on the relationship between data and software is an example of such text. Descriptions of historical incidents and accidents are another.

Each volume of the guidance document contains a single text type.

## 1.5 Conformance

There may be occasions when it is desirable or necessary to make a claim of conformance to the objectives listed in this document. Such a claim may be required, for example, if this document is explicitly included as a normative reference from a formal standard. Alternatively, it may be required as part of an organization's internal processes.

To facilitate conformance claims, the following terminology is used within the normative parts of this guidance document:

**SHALL** denotes items where evidence of conformance must be provided in order to claim conformance with this guidance document.

**SHOULD** denotes items where, in some circumstances, there may be valid reasons for not conforming to a particular item. The full implications of non-conformance must be understood, documented and approved in order to claim conformance with this guidance document.

**MAY** denotes items that are optional. These may be advantageous in some circumstances but not in others. Organizations are free to adopt any approach to these items without the need for further justification.

The terms have their normal English meanings volumes 2 and 3.

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## 2 Data Properties

*Failure is an amazing data point that tells you which direction not to go.*  
**Payal Kadakia**

[Data properties](#) are used to establish what aspects of the data (e.g., [timeliness](#), [accuracy](#)) need to be guaranteed in order that the system operates in a safe manner.

James Inge's work [2] produced a useful taxonomy of data categories, and went on to look at faults in data. He concluded that a rigid taxonomy of data categories was unhelpful due to various properties or characteristics of the data which vary independently. In short, it is the combination of data category with the required [data properties](#) that facilitates safety analysis.

Data categories were discussed Chapter 2:12. [Table 2.1](#) documents a non-exhaustive collection of [data properties](#). Typically, it is the loss of one of these [data properties](#) that presents a [hazard](#). This notion of "loss" is dependent on the intended use; for example, what is "timely" for one use may not be for another.

The "[Goldilocks](#)" property<sup>1</sup> addresses appropriate sizing and quantity of data. A number of issues have been found to arise when there is too much or too little data. While it is particularly relevant to communications links, it may have relevance to other areas such as [databases](#) and when people are involved in reviewing or checking data. The property is named "[Goldilocks](#)" as it refers to the need to have not too much, not too little, but just the right amount of data. A system where the property was lost involved a high-speed data bus that connected several safety-critical systems. A transceiver of that bus failed and transmitted random noise. The receivers employed parity checks and cyclic redundancy check, but the system had been designed to eliminate occasional [data errors](#). When random noise filled the bus, several apparently valid messages were created every second, resulting in potentially lethal behaviour.

In a [hazard and operability study \(HAZOP\)](#) carried out during 2020, based upon the [HAZOP](#) guidewords in this guidance, the facilitator realised that certain failure modes had not been identified by the [HAZOP](#) team. In addition to the issue of system overload already discussed, those omissions also concerned system behaviour following data rejection. In these cases, bad data was detected and rejected, but the consequences of data rejection over an extended period had not been considered.

The [Goldilocks](#) property is related to the data volume problem discussed in Section 3:2.2, however, given the importance of data sizing and the experience of real-world incidents this is now a separate property.

### 2.0.1 The analysability property

The [analysability](#) property recognizes that data is now highly complex and extensive, often large scale, and distributed. And yet, for safety purposes we need to make sure it is of suitable quality and able to support system goals. We therefore need to ensure that it is possible to analyse it for key characteristics and establish meaningful results using tools or other means. This property is related to explainability and may be performed by the same set of tools or techniques.

<sup>1</sup> The authors are aware that some of the names of categories are very culturally dependent. For example, we are informed that the folk-tale of Goldilocks is not well known in Chinese culture but the idiom guòyóubùjǐ, interpreted as "doing too much is just as bad as not doing enough" is well known. The intention of the names is simply to evoke the associations, so users of this guidance are encouraged to use names for the categories that are more appropriate to their cultures.



## 2.0.2 The explainability property

**Explainability** describes the ability to establish what the purpose and effect data (and especially changes to data) has on a system and explain this to relevant **stakeholders** in terms that they can understand. It is particularly important for learning and AI-based systems. This could be, for example, **ML** training data or system **configuration data**. This property is related to **analysability** and may be performed by the same set of tools or techniques.

Table 2.1: Properties of data

| Property                     | Abbreviation | Description                                                                                                               |
|------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------|
| Integrity                    | I            | The data is correct, true and unaltered                                                                                   |
| Completeness                 | C            | The data has nothing missing or lost                                                                                      |
| Consistency                  | N            | The data adheres to a common world view (e.g., units)                                                                     |
| Continuity                   | Y            | The data is continuous and regular without gaps or breaks                                                                 |
| Format                       | O            | The data is represented in a way which is readable by those that need to use it                                           |
| Accuracy                     | A            | The data has sufficient detail for its intended use                                                                       |
| Resolution                   | R            | The smallest difference between two adjacent values that can be represented in a data storage, display or transfer system |
| Traceability                 | T            | The data can be linked back to its source or derivation                                                                   |
| Timeliness                   | M            | The data is as up to date as required                                                                                     |
| Verifiability                | V            | The data can be checked and its properties demonstrated to be correct                                                     |
| Availability                 | L            | The data is accessible and usable when an authorized entity demands access                                                |
| Fidelity / representation    | F            | How well the data maps to the real-world entity it is trying to model                                                     |
| Priority                     | P            | The data is presented / transmitted / made available in the order required                                                |
| Sequencing                   | Q            | The data is preserved in the order required                                                                               |
| Intended destination / usage | U            | The data is only sent to those that should have access to it                                                              |
| Accessibility                | B            | The data is visible only to those that should see it                                                                      |
| Suppression                  | S            | The data is intended never to be used again                                                                               |
| History                      | H            | The data has an audit trail of changes                                                                                    |
| Lifetime                     | E            | When does the safety-related data expire                                                                                  |
| Disposability / deletability | D            | The data can be permanently removed when required                                                                         |
| Goldilocks                   | G            | The data is just the right size – not too much and not too little                                                         |
| Analysability                | Z            | The data (including any <b>metadata</b> ) is of a suitable size, type and format to enable it be usefully analysed        |
| Explainability               | X            | The data can be meaningfully explained, by a suitable mechanism, to those who need to understand it                       |
| Clarity                      | K            | The data is visible, unmasked, not hidden or obscured and not camouflaged                                                 |

Table 2.2 illustrates where the data issues discussed in Section 3:2.2

Table 2.2: Properties that can be lost through issues

|                                | Integrity | Completeness | Consistency | Continuity | Format | Accuracy | Resolution | Traceability | Timeliness | Verifiability | Availability | Fidelity / representation | Priority | Sequencing | Intended destination / usage | Accessibility | Suppression | History | Lifetime | Disposability / deletability | Goldlocks | Analysability | Explainability | Clarity |
|--------------------------------|-----------|--------------|-------------|------------|--------|----------|------------|--------------|------------|---------------|--------------|---------------------------|----------|------------|------------------------------|---------------|-------------|---------|----------|------------------------------|-----------|---------------|----------------|---------|
| Issue                          | I         | C            | N           | Y          | O      | A        | R          | T            | M          | V             | L            | F                         | P        | Q          | U                            | B             | S           | H       | E        | D                            | G         | Z             | X              | K       |
| Fluidity                       |           |              |             |            |        |          |            | x            |            | x             |              | x                         |          |            |                              |               |             | x       |          |                              | x         | x             | x              | x       |
| Reuse                          |           | x            |             |            |        | x        |            | x            | x          | x             |              | x                         |          |            | x                            |               | x           | x       | x        | x                            | x         |               | x              |         |
| Ageing                         |           | x            |             |            |        | x        |            |              | x          |               |              | x                         |          |            | x                            | x             | x           | x       | x        | x                            |           |               | x              |         |
| Transformation                 | x         | x            | x           | x          | x      | x        | x          | x            | x          | x             | x            | x                         | x        | x          | x                            | x             | x           | x       | x        | x                            | x         | x             | x              | x       |
| Ownership                      |           |              |             |            |        |          |            | x            |            | x             | x            |                           |          |            | x                            | x             |             | x       |          |                              |           |               | x              |         |
| Archiving / retrieval          |           |              |             |            |        |          |            |              |            |               |              |                           |          |            | x                            | x             | x           | x       | x        | x                            |           | x             |                | x       |
| Biasing                        | x         | x            | x           | x          |        | x        |            |              | x          | x             |              | x                         | x        | x          |                              |               |             |         |          |                              |           | x             | x              |         |
| Falsification / misinformation | x         | x            | x           | x          | x      | x        | x          | x            | x          | x             | x            | x                         | x        | x          | x                            | x             | x           | x       | x        | x                            | x         | x             | x              | x       |
| Defaulting                     | x         | x            | x           | x          |        | x        |            |              |            |               |              | x                         |          |            |                              |               |             |         |          |                              |           | x             |                | x       |
| Sentinels                      | x         | x            |             | x          |        | x        |            |              |            |               |              |                           |          | x          |                              |               |             |         |          |                              |           | x             |                | x       |
| Aliasing                       | x         | x            |             | x          |        | x        | x          | x            |            | x             | x            | x                         |          |            |                              |               |             |         |          |                              |           | x             | x              | x       |
| Disassociation                 | x         |              |             | x          |        | x        |            | x            | x          | x             | x            | x                         | x        | x          | x                            | x             | x           | x       | x        | x                            |           | x             | x              | x       |
| Masking                        | x         | x            | x           | x          |        | x        |            |              |            | x             |              | x                         |          |            |                              |               |             |         |          |                              |           | x             | x              | x       |
| Incompleteness                 | x         | x            | x           | x          | x      | x        | x          | x            | x          | x             | x            | x                         | x        | x          | x                            | x             | x           | x       | x        | x                            | x         | x             | x              | x       |
| Volume                         |           |              |             |            |        |          |            |              |            | x             |              |                           |          |            |                              |               |             |         |          |                              | x         | x             | x              | x       |
| Interpretation                 | x         | x            | x           |            | x      | x        | x          |              | x          | x             |              | x                         |          |            |                              |               |             |         |          |                              |           | x             | x              |         |
| Distribution                   | x         | x            | x           | x          |        |          |            | x            |            |               | x            |                           | x        | x          | x                            | x             |             | x       |          | x                            |           | x             | x              | x       |

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## 3 Principles

*Errors using inadequate data are much less than those using no data at all.*  
**Charles Babbage**

Hawkins *et al.* established some generic software safety assurance principles, which are commonly referred to as “4 + 1” [3]. Given the close links between software and data it is helpful to consider these principles from a data safety assurance perspective. The results are detailed below, with each principle being considered in turn.

### 3.1 Principle 1: data safety requirements shall be defined to address the data contribution to system hazards

Data pervades active system operation, as well as the system’s specification, realisation, [verification](#), [validation](#), certification, training, maintenance, and retirement. Moreover, data may be passed from one system to another, sometimes over a significant period of time. Data may be assimilated and converted from prior uses into new uses, or simply used as-is, by many systems. It is stored in media whose storage [integrity](#) decays. The system context for [data safety requirements](#) may be specific to a particular system’s or process’s use of the data, or it may be generalised to a class of related systems. Hence [data safety requirements](#) are needed for any safety-related system that interacts with data.

### 3.2 Principle 2: the intent of the data safety requirements shall be maintained throughout requirements decomposition

[Data safety requirements](#) establish the system’s safety properties for data, for the system’s use of data, for the management of data, and for the engineering lifecycle of both the system and its associated data. The system’s requirements hierarchy must preserve the intent of the [data safety requirements](#) (and hence the system’s safety-related [data properties](#)). Moreover, the applied engineering process for both the system’s realisation and subsequent lifecycle stages shall demonstrate that the data safety properties are preserved.

### 3.3 Principle 3: data safety requirements shall be satisfied

Evidence is required that the system satisfies all of the [data safety requirements](#) imposed on it for all anticipated operating conditions. Moreover, the [data safety requirements](#) that pertain to the data’s lifecycle outside of the system shall be evidentially demonstrated prior to the system acting on such data, or the system shall be able to adequately defend against unsatisfied [data safety requirements](#). In other words, either the data can be shown to demonstrate the required [data properties](#) prior to being used or the system can implement adequate defences and [mitigations](#) against data that does not conform to the required safety properties. Sufficient resilience against unknowns, e.g. dark data, needs to be demonstrated.

### 3.4 Principle 4: Hazardous system behaviour arising from the system's use of data shall be identified and mitigated

This is an intentionally broad statement because data is conceptual and not physical; it is the contextualised use of data that could result in a system hazard. Data safety assurance principle 1 deals with system-level hazards arising from data, whereas Data safety assurance principle 4 is concerned with hazards that arise from the way the system uses its data, that is, whether the system's design and implementation introduce further hazards. An example is a ship navigation system's display of hydrographic chart data, where a wide field display results in small features disappearing (due to image scale) when it is critical that situational awareness of such hazards is maintained. The associated objectives and outputs are shown in Table ??.

### 3.5 Principle 5: The confidence established in addressing the data safety assurance principles shall be commensurate to the contribution of the data to system risk

(Called Principle 4+1 by Hawkins.) The confidence in the evidence that demonstrates establishment of the first four Data Safety Assurance Principles shall be proportionate to the contribution data (or a particular [data artefact](#)) makes to the system hazards.

### 3.6 Principle 6: Principle 6: All principles SHALL be established and maintained through all changes, reuse, re-deployment and re-purposing.

It is recognised that systems involving data are always subject to change. This could be change to the system or the data involved, or may be a change of use. Data may grow in size, evolve or change in many ways e.g. increased or decreased accuracy from sensor data. A new use of the system could impose different constraints on data properties and require production of new or updated evidence. This means that the principles and objectives may have to be reviewed after any change. A process to detect and review changes needs to be in place.

## 4 Definitions

*I love data. I think it's very important to get it right, and I think it's good to question it.*

**Mary Meeker**

### **artefact, data**

An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.

### **property, data**

A characteristic that can be exhibited by a [data artefact](#).

### **safety assurance level, data**

An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

### **safety requirement, data**

A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

### **stakeholder**

An individual or organization that has some relationship to the system, possibly including a power of veto.

### **treatment**

An action taken to reduce or control risk. This might be [mitigation](#) of the risk or elimination of the hazard.

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## 5 Acronyms, Definitions and Glossary

*The plural of anecdote is not data.*  
**Mark Berkoff**

### 5.1 Acronyms

|        |                                             |
|--------|---------------------------------------------|
| AI     | artificial intelligence                     |
| CONOPS | Concept of Operation                        |
| DSAL   | <a href="#">data safety assurance level</a> |
| DSG    | data safety guidance                        |
| DSIWG  | Data Safety Initiative Working Group        |
| HAZOP  | hazard and operability study                |
| LLM    | large language model                        |
| ML     | machine learning                            |
| OED    | Oxford English Dictionary                   |
| SCSC   | Safety-Critical Systems Club                |
| SSS    | Safety-critical Systems Symposium           |

### 5.2 Definitions and Glossary

#### **accessibility**

Property that the data is visible only to those that should see it.

#### **accuracy**

Closeness of agreement between a measured quantity value and a true quantity value of a measurand. [4]

#### **analysability**

The data (including any [metadata](#)) is of a suitable size, type and format to enable it be usefully analysed

#### **artefact, data**

An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.



**availability**

The property of being accessible and usable upon demand by an authorized entity. ISO 27001:2013 [5]

**clarity**

The property that The data is visible, unmasked, not hidden or obscured, and not camouflaged

**completeness**

- Property of having every necessary part or element.
- Completeness of the data provided. RTCA/DO-200A

**configuration data**

- Data that configures a generic software system to a particular instance of its use. (EC) No 482/2008 [6]
- Data that configures a generic software system to a particular instance of its use (e.g., data for flight data processing system for a particular airspace, by setting the positions of airways, reporting points, navigation aids, airports and other elements important to air navigation). ED-153 [7] Data that configures a generic software system to a particular instance of its use (e.g., data for flight data processing system for a particular airspace, by setting the positions of airways, reporting points, navigation aids, airports and other elements important to air navigation). ED-153 [7]

**consistency, data**

The property that the data adheres to a common world view (e.g., units).

**continuity, data**

The property that the data is continuous and regular without gaps or breaks.

**correctness, data**

self-consistency, protection against alteration or corruption and consistency with the functional requirements of the data-driven system. IEC 61508 Part 3 [8]

**data**

- A thing given or granted; something known or assumed as fact, and made the basis of reasoning or calculation; an assumption or premiss from which inferences are drawn. OED
- A reinterpretable representation of information in a formalized manner suitable for communication, interpretation or processing. ISO/IEC 2382 [9]

**database**

A set of data, part or the whole of another set of data, consisting of at least one file that is sufficient for a given purpose or for a given data processing system. RTCA/DO-178C

**data-driven system**

System which relies upon configuration data or lookup tables to define the functionality of the system. IEC 61508 Part 4 [10]

**dataset**

Identifiable collection of data. Note that a dataset may be a smaller grouping of data which, though limited by some constraint such as spatial extent or feature type, is located physically within a larger dataset. Theoretically, a dataset may be as small as a single feature or feature attribute contained within a larger dataset. A hardcopy map or chart may be considered a dataset. BS EN ISO 19131:2008 [11]

**disposability / deletability**

The property that the data can be permanently removed when required

**error, data**

- Discrepancy with the universe of discourse. ISO 19138:2006 [12]
- Discrepancy between a data value and the true, specified or theoretically correct value or condition. P. Ensor [13]

**explainability**

The property that the data can be meaningfully explained, by a suitable mechanism, to those who need to understand it.

**fidelity / representation ,data**

The property describing how well the data maps to the real world entity it is trying to model.

**format**

The property that data is represented in a which is readable by those that need to use it.

**Goldilocks**

The property that the data is just the right size – not too much and not too little

**hazard, data**

Use of data (in the context of a system) that could lead to harm. SCSC DSIWG

**history**

Property that the data has an audit trail of changes.

**information**

- Knowledge communicated concerning some particular fact, subject, or event; that of which one is apprised or told - intelligence, news - as contrasted with data. OED
- Knowledge that has a contextual meaning. ISO/IEC 2382 [9]

**integrity, data**

- The assurance that a data element retrieved from a storage system has not been corrupted or altered in any ways since the original data entry or latest authorised amendment. RTCA/DO-200A [14]
- The degree of assurance that a [data item](#) and its value have not been lost or altered since the data origination or authorised amendment. (EU) No 73/2010 [15]
- The degree of undetected (at system level) non-conformity of the input value of the [data item](#) with its output value. (EU) No 1207/2011 [16]

- The property of protecting the [accuracy](#) and [completeness](#) of assets, i.e., that which has value to the organization. ISO 27001:2013 [5]

### **intended destination / usage**

Property that the data is only sent to those that should have access to it

### **item, data**

Single attribute of a complete [dataset](#), which is allocated a value that defines its current status. (EU) No 73/2010 [15]

### **lifetime**

The property of when the safety-related data expire

### **metadata**

Data that represents [information](#) about data itself. Note that one should distinguish between “Structural Metadata”, which is data about the design and specification of data structures (and is more properly called “data about the containers of data”) and “Descriptive Metadata”, which is about individual instances of application data, the data content. J. Inge [2]

### **mitigation**

Steps taken to control or prevent a hazard from causing harm and reduce risk to a tolerable or acceptable level. [17]

### **owner, data**

The individual or organization responsible for a particular [data artefact](#) or collection of [data artefacts](#).

### **priority**

The property that data is presented / transmitted / made available in the order required.

### **property, data**

A characteristic that can be exhibited by a [data artefact](#).

### **resolution**

- The ability of a device to respond to small differences in input and to indicate or represent them accurately in output; a measure of this, expressed as the smallest difference so distinguishable. OED.
- The smallest difference between two adjacent values that can be represented in a data storage, display or transfer system. RTCA/DO-200A [14]
- The number of units or digits to which a measured or calculated value is expressed and used. (EU) No 73/2010 [15]

### **response**

The way in which an identified risk is addressed; possible responses include avoid / eliminate, treat, or accept as sufficiently low.

### **safety**

- The state of being protected from or guarded against hurt or injury; freedom from danger. ??

- Freedom from conditions that can cause death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment. [1]

### **safety assessment, data**

The process of explicitly considering data as part of a system safety assessment, via the means of [data artefacts](#), Data Properties and [DSALs](#).

### **safety assurance level, data**

An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

### **safety requirement, data**

A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

### **sequencing**

The property that the data is preserved in the order required.

### **stakeholder**

An individual or organization that has some relationship to the system, possibly including a power of veto.

### **suppression**

Property that the data is intended never to be used again

### **tailoring**

Adaptation of processes etc. to be appropriate for a specific system and context

### **timeliness**

- A measure of the time delay between a change in the real world and the associated [database](#) update being available to the user. P. Ensor [13]
- The difference between the time of output of a [data item](#) and the time of applicability of that [data item](#). (EU) No 1207/2011 [16]

### **traceability**

Ability to determine the origin of the data. RTCA/DO-200A [14]

### **treat**

To apply a [treatment](#)

### **treatment**

An action taken to reduce or control risk.

### **validation, data**

- The activity whereby a data element is checked as having a value that is fully applicable to the identity given to the data element, or a set of data elements that is checked as being acceptable for their purpose. RTCA/DO-200A [14]
- Process of ensuring that data meets the requirements for the specified application or intended use. (EU) No 73/2010 [15]

**verifiability**

Evaluation of the output of an [aeronautical data](#) process to ensure [correctness](#) and [consistency](#) with respect to the inputs and applicable data standards, rules and conventions used in that process. (EU) No 73/2010 [[15](#)]

**verification, data**

Evaluation of the output of a process to ensure [correctness](#) and [consistency](#) with respect to the inputs and applicable data standards, rules and conventions used in that process. Based on (EU) No 73/2010 [[15](#)]

The normative list of definitions is at [chapter 4](#). Normative definitions have been repeated here for convenience.

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*Data opportunities multiply as the data is transformed.*  
**Sun Tzu misquoted**

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## **Data is here. Data is Growing. Data is Causing Harm**

**These books have been developed by the Safety-Critical Systems Club Data Safety Initiative Working Group (DSIWG) to provide guidance on how data, as distinct from hardware and software, can be managed in a safety-related context.**

**"If you torture the data long enough, they confess – even to crimes that were never committed.**

**Nihat Bülent Gültekin**



**This is a major rewrite of the version 3 guidance, to improve the structure, clarify the workflow, improve the consistency of the normative content and to move much of the appendix material of version 3 into the body of the text. Although the fundamental process has not changed, conformance to previous versions does not guarantee conformance to this version.**